

Batch Verification for Statistical Zero Knowledge

Done non-interactively; the interactive class is the “most pressing open question”

Status. Statement: AI-written from Changrui Mu, Shafik Nassar, Ron D. Rothblum and Prashant Nalini Vasudevan, *Strong Batching for Non-Interactive Statistical Zero-Knowledge*, IACR ePrint 2024/229. Not yet checked by a human. Their Theorem 1.1 gives a batch NISZK proof with communication $\text{poly}(n, \log k)$ for k instances; they name the SZK analogue as “[t]he most pressing open question” in their discussion. A $\text{poly}(n)$ dependence is unavoidable even at $k = 1$, so $\log k$ is the right target in the second parameter.

Category. *Zero Knowledge.*

Conjecture (informal). *Proving k statements should cost less than proving each one separately. For zero-knowledge proofs this is delicate, because the obvious savings — reuse randomness, combine instances — are exactly the moves that threaten the zero-knowledge property. For the non-interactive class NISZK the source settles it: k instances can be batched into a single proof whose communication grows only logarithmically in k . For the interactive class SZK, which is larger and where the analogous protocols are the older and better studied ones, nobody knows. The source calls it the most pressing question its work leaves.*

1 The setting

Write $\Pi^{\otimes k}$ for the set of k -tuples of equal-length instances all of which belong to Π . Batch verification asks for a proof for $\Pi^{\otimes k}$ substantially cheaper than k independent executions, while preserving the proof system’s guarantees — here, statistical zero knowledge.

What the source proves is the non-interactive case, in a strong form.

Theorem 1 (Batch proofs for NISZK, [MNRV24, Theorem 1.1]).

Suppose $\Pi \in \text{NISZK}$ and $k = k(n) \in \mathbb{N}$ with $k(n) \leq 2^{n^{0.01}}$, where n is the length of a single instance of Π . Then $\Pi^{\otimes k}$ has an NISZK protocol in which the communication complexity and the length of the common random string are $\text{poly}(n, \log k)$. The completeness, soundness and zero-knowledge errors are all negligible in n and k , and the verifier runs in time $\text{poly}(n, k)$.

Two things about the shape of that bound. The source records that “a $\text{poly}(n)$ dependence in the communication complexity is inevitable, even when $k = 1$, assuming the existence of a sub-exponentially hard problem in NISZK”, which follows from known limitations on laconic provers. So $\text{poly}(n)$ is not slack, and $\log k$ is the aggressive part. And the protocol is obtained by reducing to NISZK-complete problems, which matters for Remark 4.

2 Notation and parameters

- n is the length of a single instance; $k = k(n)$ the number of instances batched.
- $\Pi^{\otimes k}$ is the k -fold conjunction: tuples all of whose components are in Π .
- SZK is the class with statistical zero-knowledge *interactive* proofs; NISZK its non-interactive counterpart, with a common random string. $\text{NISZK} \subseteq \text{SZK}$.
- The target is total communication; the source’s NISZK result also bounds the common random string.

3 The conjecture

Conjecture 1 ([MNRV24, §1.3, item 1]). *For every $\Pi \in \text{SZK}$ there exists an SZK proof for $\Pi^{\otimes k}$ with communication $\text{poly}(n, \log k)$.*

Remark 1 (How the source states it, and the weaker fallback it offers). Page 12: “The most pressing open question is whether a similar result holds for SZK – namely, for every $\Pi \in \text{SZK}$ does there exist an SZK proof for $\Pi^{\otimes k}$ with communication $\text{poly}(n, \log k)$? Or, alternatively, one that features less stringent, yet non-trivial, communication such as sub-linear dependence on k ?”

Two targets, and the second is much weaker: *any* sub-linear dependence on k would already be non-trivial. A resolution should say which it achieves. Conjecture 1 records the strong form because that is the one the source names first and the one matching what it proved for NISZK; a sub-linear-in- k result is genuine progress and should be reported as partial rather than as a settlement.

Remark 2 (Why the non-interactive proof does not obviously transfer). The source’s route is specific and its steps are what a SZK argument would have to replace. It reduces the k instances to k instances of an NISZK-complete problem, then reduces those to a single instance of another complete problem, using two technical observations: that hash functions with bounded independence — 4-wise suffices — preserve very specific types of entropies, and that a cascade of such hash functions can be derandomized while still preserving that behaviour.

Both steps lean on the structure of the NISZK-complete problems, which are entropy-approximation problems. SZK has its own complete problems (statistical difference), and whether the same entropy-preserving and derandomization machinery applies to them is precisely what is not known.

Remark 3 (What is known on the interactive side already). This is not a question with no prior work: the source positions itself against a line including Kaslasi et al. and Kaslasi, Rothblum and Vasudevan on batch verification, and notes neighbouring results on zero-knowledge proofs for conjunctions and on computationally sound protocols for monotone policies within NP. What is absent is a $\text{poly}(n, \log k)$ bound for all of SZK.

Remark 4 (Neighbouring questions in the same list, none of them this one). The source lists two more. First, on *prover efficiency*: problems in $\text{SZK} \cap \text{NP}$ have SZK protocols with an efficient prover given an NP-witness, but all current batch protocols proceed by reducing to complete problems and so do not preserve that efficiency; is there a batch protocol even for $\text{NISZK} \cap \text{NP}$ that does? Second, on *rate*: can the multiplicative overhead be a fixed constant, i.e. communication $O(c) + \text{polylog}(n, k)$ where c is the communication for a single instance, and can that constant be pushed toward 1? The source notes this “might necessitate avoiding the complete problems, since the reduction introduces a polynomial overhead”.

Both are consequences of the same reduction-to-complete-problems route as Remark 2, which is why that route is the thing to attack.

Remark 5 (Bounds on k are part of the statement). Theorem 1 carries the hypothesis $k(n) \leq 2^{n^{0.01}}$. Conjecture 1 as the source phrases it names no range for k ; a resolution should state the range it covers, since a result for k polynomial in n is weaker than one matching the NISZK theorem’s reach.

4 Bibliography

- [MNRV24] Changrui Mu, Shafik Nassar, Ron D. Rothblum and Prashant Nalini Vasudevan. *Strong batching for non-interactive statistical zero-knowledge*. IACR ePrint 2024/229. The source. Theorem 1.1 is on page 4 and re-proved on page 20; the discussion and open problems are Section 1.3 on page 12.
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- [GH98] Oded Goldreich and Johan Håstad. *On the complexity of interactive proofs with bounded communication*. Information Processing Letters, 1998. The other.
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- [KRR+20] Inbar Kaslasi, Guy N. Rothblum, Ron D. Rothblum, Adam Sealfon and Prashant Nalini Vasudevan. *Batch verification for statistical zero knowledge proofs*. TCC 2020. The prior batch-verification protocols the source simplifies and strengthens. [UNVERIFIED] — cited in the source as [KRR+ 20]; the bibliographic detail here is inferred.